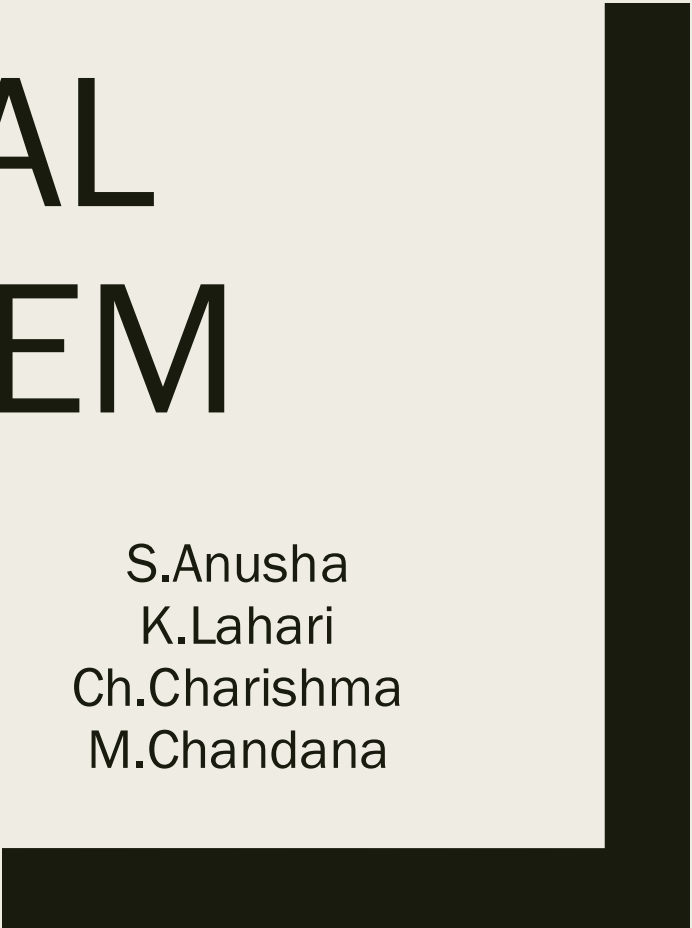




TRAFFIC SIGNAL CONTROL SYSTEM

S.Anusha
K.Lahari
Ch.Charishma
M.Chandana



ABSTRACT

- Traffic congestion is a common problem in urban areas. Efficient traffic management is necessary to reduce waiting time and ensure smooth vehicle flow. This project presents an Intelligent Traffic Control System using the Queue Data Structure (FIFO – First In First Out).
- The system simulates three traffic lanes where vehicles are managed dynamically. Special priority is given to emergency vehicles like ambulances. If an ambulance is present, it is immediately allowed to pass by giving a green signal to that lane. Otherwise, the system selects the lane with the highest number of vehicles.
- The system runs in cycles, allowing one vehicle to pass per cycle. This improves fairness and efficiency. The project demonstrates how data structures like queues can be applied to solve real-world problems.

INTRODUCTION

- Traffic management is an important aspect of smart cities. Traditional traffic systems use fixed timers, which are not efficient during varying traffic conditions.
- This project introduces a smarter approach using queues:
- Each lane is treated as a queue.
- Vehicles enter at the rear and exit from the front.
- The system dynamically decides which lane gets the green signal.
- Key features:
- Fair vehicle movement using FIFO principle
- Priority handling for ambulances
- Dynamic lane selection based on traffic density
- This approach reduces waiting time and improves emergency response.

ALGORITHM

Step 1:START

Step 2:DEFINE lanes:

- Lane1, Lane2, Lane3

Step 3:INITIALIZE:

- All lanes as empty
- cycle = 0
- maxCycles

Step 4: ADD VEHICLES

Insert vehicles into each lane

Step 5: DISPLAY LANES

1. For each lane:
 - a. Print all vehicles in order

Step 6:CHECK EMERGENCY (AMBULANCE)

1. For each lane:
 - a. Traverse the lane
 - b. If "Ambulance" is found:
 - i. Remove it from its position
 - ii. Insert it at the front of the same lane
 - iii. Mark this lane as priority
 - iv. Stop checking further lanes

Step 7: SELECT LANE FOR GREEN SIGNAL

1. If any lane has ambulance:
 - a. Select that lane
2. Else:
 - a. Compare sizes of all lanes
 - b. Select lane with maximum vehicles

Step 8: MOVE VEHICLE (SIGNAL GREEN)

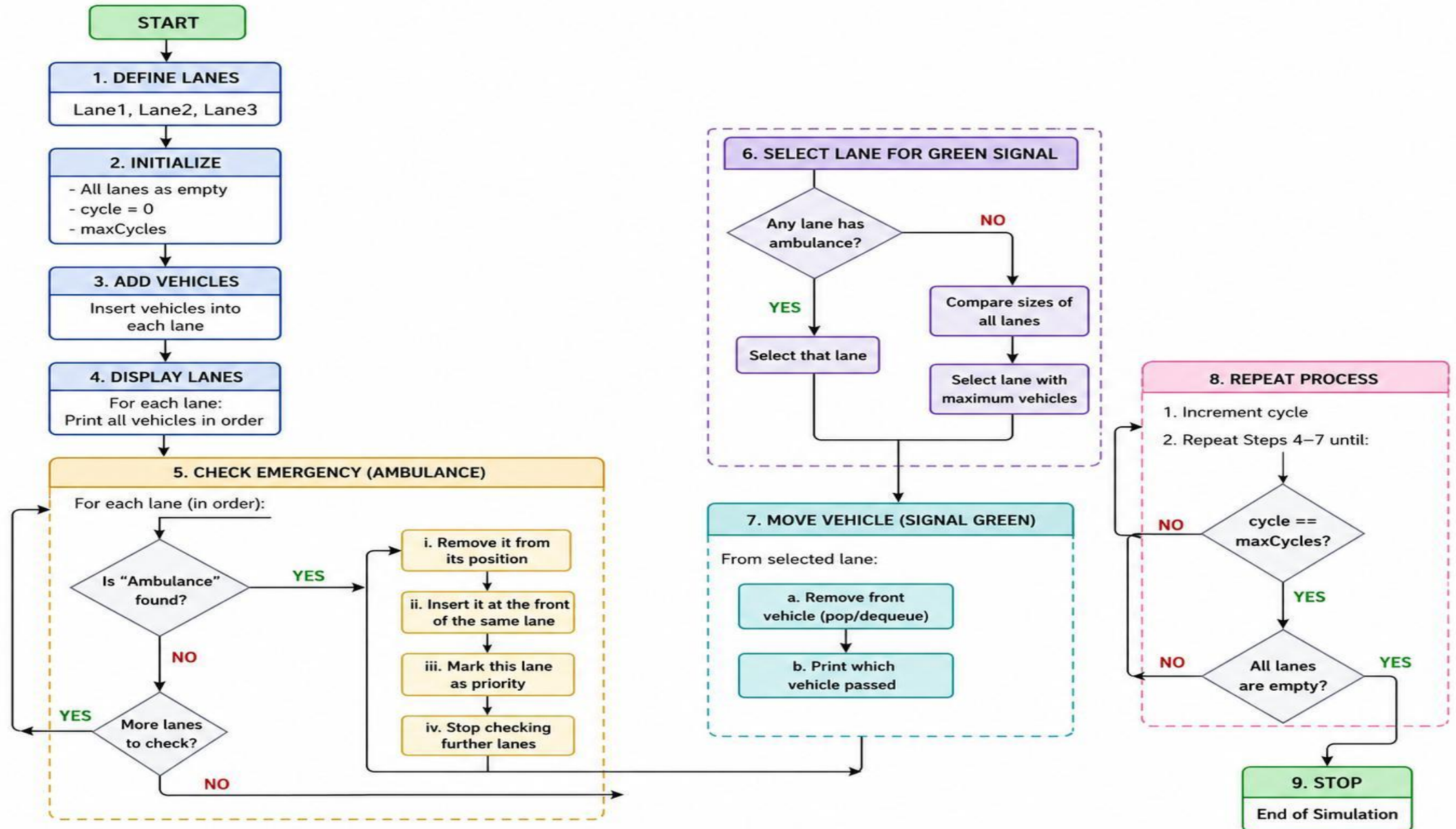
1. From selected lane:
 - a. Remove front vehicle (pop/dequeue)
 - b. Print which vehicle passed

Step 9: REPEAT PROCESS

1. Increment cycle
2. Repeat Steps 5–8 until:
 - a. Cycle == maxCycles OR
 - b. All lanes are empty

Step 10:STOP.

FLOW CHART



METHODOLOGY

The main ideology behind this system is to create an efficient, intelligent, and emergency-priority traffic control mechanism using queues and dynamic signal selection.

Traffic is organized into multiple lanes where each lane behaves like a queue following the FIFO (First In First Out) principle. Vehicles move in the same order in which they arrive, ensuring fairness and maintaining discipline in traffic flow.

The system gives special priority to emergency vehicles such as ambulances. When an ambulance is detected in any lane, it is moved to the front of that queue. The corresponding lane is immediately given the green signal to allow quick passage, reducing delays and helping in critical situations.

In normal conditions where no emergency vehicle is present, the system dynamically selects the lane with the highest number of vehicles. This ensures that traffic congestion is reduced by clearing the busiest lane first.

The system continuously monitors all lanes in repeated cycles. Based on real-time traffic conditions, it updates the signals dynamically instead of relying on fixed timers.

Queue operations such as enqueue and dequeue are used to manage vehicle movement efficiently. This ensures smooth processing with minimal complexity.

Overall, the system automates traffic signal decisions, improves road efficiency, reduces waiting time, and provides a smarter and safer traffic management solution.

RESULT(ScreenShot)

```
C:\Users\ANUSHA\OneDrive\Desktop\drcode.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
[global]
Project Classes Debug dproject2.cpp x drcode.cpp x
99 return true;
100
101 3. Display Traffic
102 4. Exit
103 Enter choice: 1
104 Select Lane (1-A, 2-B, 3-C, 4-D): 1
105 Enter vehicle type: car
106
107 ===== TRAFFIC MENU =====
108 1. Add Vehicle
109 2. Run Signal Cycle
110 3. Display Traffic
111 4. Exit
112 Enter choice: 1
113 Select Lane (1-A, 2-B, 3-C, 4-D): 1
114 Enter vehicle type: bike
115
116 ===== TRAFFIC MENU =====
117 1. Add Vehicle
118 2. Run Signal Cycle
119 3. Display Traffic
120 4. Exit
121 Enter choice: 1
122 Select Lane (1-A, 2-B, 3-C, 4-D): 2
123 Enter vehicle type: bike
124
125 ===== TRAFFIC MENU =====
126 1. Add Vehicle
127 2. Run Signal Cycle
128 3. Display Traffic
129 4. Exit
130 Enter choice:
Compiler (v) Resources
Abort Compilation
Shorten compiler paths
Line: 281 Col: 2 Sel: 0 Lines: 281 Length: 6746 Insert Done parsing in 0.051 seconds
```

CONCLUSION

- The Intelligent Traffic Control System using queues provides an efficient and practical solution for managing traffic.
- Key advantages:
- Uses simple data structures (queue)
- Ensures fairness using FIFO
- Handles emergency vehicles effectively
- Reduces congestion by selecting busiest lane
- This project demonstrates how theoretical concepts like data structures can be applied to real-world problems like traffic management.

FUTURE SCOPE

- Future Scope
- This system can be improved further with advanced technologies:
- AI-Based Traffic Prediction
- Predict traffic using machine learning
- Sensor-Based Vehicle Detection
- Use cameras and IoT sensors for real-time data
- Multiple Emergency Handling
- Handle multiple ambulances simultaneously
- Real-Time Traffic Integration
- Connect with smart city infrastructure
- Mobile App Integration
- Monitor traffic remotely